An Evaluation of SPC Day-1 Tornado Probabilities from the Public's Perspective for Aurora, IL (2015–2025)

By Ben Zusman

Jan 5, 2026

Independent research project (no institutional affiliation)

Note: This project is exploratory and does not particularly reflect a comprehensible forecast calibration from a meteorological verification standpoint. Instead, it is based on SPC Day-1 Tornado probabilities as they are publicly communicated. This analysis focuses on whether the observed tornado frequencies are consistent with the stated SPC probabilities under the statistical assumptions that are described below. Because this study uses a single local point and a finite sample, the results should not be interpreted as a measure of overall SPC forecast accuracy or national forecast performance.

Abstract

This exploratory local calibration study evaluates the Storm Prediction Center's Day-1 tornado probabilities based on what the user would experience in the time span of 2015-2025. As such, one point is chosen to represent an average user, and statistical tests are done according to the available public communication of the SPC forecasts. As SPC forecasts are made such that the public can understand their risk, it is important that it is accurate in the way that the public would interpret it. While national meteorological evaluations of SPC's calibration exist, a local evaluation from the public's standpoint is a new question. The main portion of this study included comparing SPC's stated tornado probabilities with observed probabilities using statistical tests such as binomial tests and Wilson confidence intervals. Typically, these tests would not be used in meteorological forecast evaluations; however, it makes sense to use them given that the SPC frames their percentages as independent to be taken at face value. Additional sensitivity tests were done using a different radius and threshold values. The observed findings were much higher than the reported percentages given by the SPC. This was particularly prevalent in the 2020-2025 time period. This result stayed true under more conservative threshold tests, while the 10-mile sensitivity was closer to accurate calibration. The SPC did a good job at identifying which days would be higher risk, having high risk stratification despite apparent calibration issues. Overall, the results suggest that, for this specific location and methodology, SPC probabilities may not be well calibrated when interpreted literally by the public, despite retaining useful information about risk stratification. However, this is a

single-location exploratory study with a small sample size, and thus does not particularly indicate findings of the SPC at large.

Introduction and Background:

The Storm Prediction Center (SPC) is a branch of the NWS (National Weather Service) and the NOAA (National Oceanic and Atmospheric Administration) government agencies that aims to inform the public by providing probabilistic forecasts for the risk of tornadoes and other severe weather hazards. The general public relies on the SPC to understand the estimated tornado risk during a certain time period and make plans accordingly.

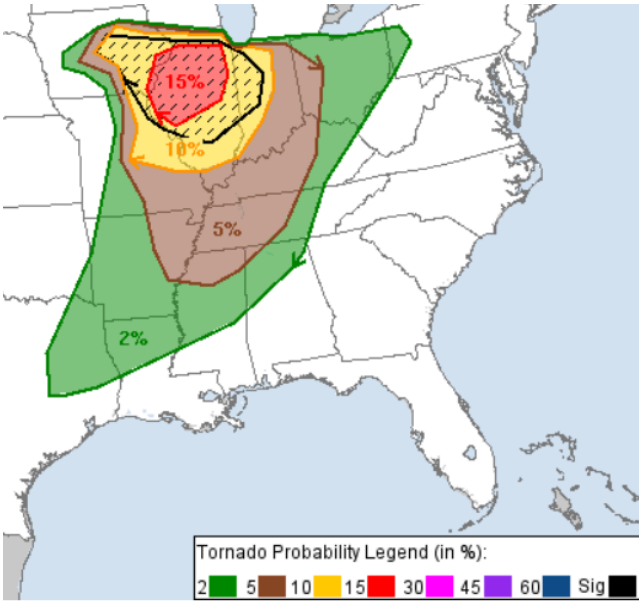
On any given day, the SPC releases outlooks (forecasts) for various weather hazards, including tornadoes. These outlooks come equipped with maps where shaded regions represent forecast probabilities in percentages, indicating the "Probability of a tornado within 25 miles of a point" (*SPC Convective Outlooks: Days 1–8*, 2026)

A tornado counts towards verification as long as it is observed and within a 25-mile radius of any given point in the risk contour; the tornado can be of any duration or size. From the SPC itself: "As an example, if you have a 15% probability for tornadoes, this means you have a 15% chance of a tornado occurring within 25 miles of your location" (Bunting & Grams, 2020).

The SPC updates these predictions at least daily as the predicted event approaches. Though the SPC tracks overall 'severe weather' risk up to 8 days out, it is on the day one (day of) and day two outlook that the SPC emphasizes forecasts for the probability of individual hazards, including tornadoes, wind, and hail. This study will focus on the Day 1 tornado outlooks, which are first released in the morning of the event day. The SPC updates its forecasts multiple times during the day of the event because the weather can rapidly change and evolve, and the risk percentages need to keep pace.

These outlooks also include a list of populated cities for each percent range, as well as a written description of possible threats in both general summary and scientific terms. In addition to regular percentages, the SPC releases 'hatched' risks, which indicate the heightened probability of a strong tornado if a tornado does occur. This study focuses on the regular tornado percentages, as there is a larger sample size of non-strong tornadoes. An example is shown in Figure 1.

Figure 1. Example of SPC tornado outlook contours.



(*Storm Prediction Center Product & Report Archives*, 2020)

As a resource the public relies on to make their lives safer and more alert, it is essential that the SPC is accurate in its tornado predictions. For an outlook to be considered well calibrated, ideally, the stated risk should match long-term observed frequency data. As in, if a 5% risk is issued across 100 independent cases, for example, one should expect a tornado to occur in the radius about 5 times (though there would be some variance in practice).

It is important that the SPC does not overestimate, as that would cause false alarms and significantly lower institutional trust if prevalent. High risks disrupt schedules, such as missing work to avoid travel during the storm, and schools or even offices have been known to give days off on high-risk days entirely dictated by these SPC predictions, so if the SPC overestimates, this would fuel frustrations towards the organization and lower future trust. It is also important that there is no underestimation of the odds, as that would also lower institutional trust. Underforecasting is still a major issue, as it can cause people to underestimate their risk and not take it as seriously. When forecasts are under a certain percentage risk, many users may neglect or essentially ignore the risk, not taking proper precautions, and be caught off guard.

At the most basic level, the SPC seems to prioritize national accuracy. To test these results, various national studies have been conducted to determine national-scale calibration. Some of the national-scale research, such as that of Gregory R. Herman, Erik R. Nielsen, and Russ S. Schumacher, suggests that “day 1 and day 2–3 tornado outlooks consistently exhibit an underforecast bias.” (Schumacher et al., 2018). Meaning that the SPC’s percent predictions are lower than the observed outcome. If the SPC is found to have an underforecast bias at the national level, questions arise about whether the underforecast bias is uniformly true or driven by regional heterogeneity in forecast performance.

While there are many verifications of the SPC, nearly all focus on national-level verification and overall trends in particular. However, the average user is not using the SPC in this way; instead, they would focus all of their attention on their specific location, as that is the only location relevant to them. As forecasts are experienced at the local level, specific local calibration is an important yet less studied question. Additionally, most focus on how well-calibrated the SPC is internally, while this study focuses on how well-calibrated the SPC is specifically in how it is framed to the public.

In professional meteorological verification, there are additional complexities and nuances. Typically, different statistical tests are used specifically for weather forecast verification; however, this study deliberately evaluates the probabilistic meaning of SPC forecasts as presented to end users, rather than attempting to reproduce a comprehensive operational forecast-verification framework. The reality is that many of the nuances involved in forecasting are not made evident to the public. However, as a government agency made to aid the public, the forecasts should still be able to provide valuable information as conveyed.

The SPC conveys the probabilities to the public in a straightforward way, and it can be reasonably interpreted that each day is statistically independent from the public's standpoint. This means that the study would treat each day as an independent Bernoulli trial, as that is how the public would interpret the forecast through the SPC's communication. As quoted earlier, 15% means 15% according to their public communication. Given that, this study will deliberately take their wording at face value, using independent statistical trials and raw percentages.

This study evaluates whether these predictions hold up at a local level from the public's perspective. National-scale calibration does not necessarily completely correspond to local calibration, so testing the latter may be useful. Overall, my goal is to analyze whether SPC Day-1 tornado probabilities are well-calibrated in the way that they are publicly expressed in this region and whether calibration has changed over time. This is not a strict meteorological accuracy study, but instead a study of how well the SPC's presentation and language can hold up from the public's perspective.

This study uses Aurora, Illinois, as the local assessment point. Aurora was chosen because it is a major population center, being the 2nd most populous city in Illinois (Carney, 2026). Aurora was selected as a large population center within a tornado-prone region of the United States, making it a relevant case for evaluating how SPC forecasts are experienced at the local level. Given that the 25-mile forecast radius around Aurora encompasses over 550,000 people, accurate local calibration has direct implications for public risk perception and emergency preparedness. The choice of having it be a local calibration was made because that is how the SPC will be used from the public's perspective.

This study spans the timeframe of 2015-2025 (including 2015 and 2025; so 11 years total), as this is both a substantial period for studying and recent enough that the SPC remains similar in practice. The SPC, particularly before 2010, has noticeable differences in methods, so 2015 can be chosen as a cut-off to keep it both modern and substantial. This timeframe was chosen to go to the present so it can be more relevant. The timeframe will mainly split the data into two sets: 2015-2019 and 2020-2025. These years were chosen because they are roughly the timeframe split in half, mark the turn of the decade, and can be used as a natural breaking point.

This study asks the following questions: (1) Were local SPC Day-1 probabilities well-calibrated at Aurora, IL from the public's perspective? (2) Did the observed relationship between SPC probability categories and tornado occurrence differ between the two periods?

Data and Methods:

A. Data sources:

Data for SPC forecasts and tornado verification were obtained from publicly available government and scientific databases.

SPC Day-1 convective outlook archive (*Storm Prediction Center Product & Report Archives*, n.d.)

SPC/NWS Detailed Storm Reports (*Storm Reports Page*, n.d.)

Google Earth (*Google Earth*, n.d.)

NOAA Lat/Long Distance Calculator (*Latitude/Longitude Distance Calculator*, n.d.)

Date range: 2015–01–01 through 2025–12–31

Reference point: Aurora, IL (lat lon: 41.75°N, 88.31°W). Aurora was chosen for being a population center. A fixed point in the downtown (City of Aurora, IL, Government building at 44 E Downer Pl, Aurora, IL 60505) was chosen to standardize calculations.

B. Selection Rules:

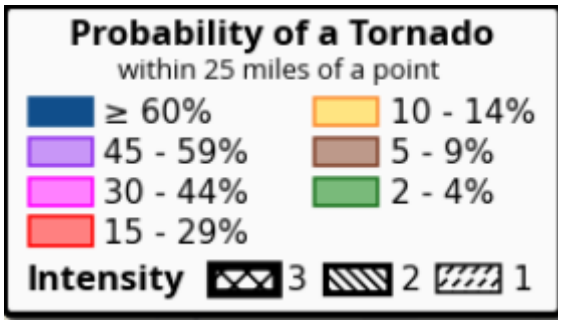
Each trial represents a single forecast day at a fixed location, evaluated using the most recent SPC outlook valid prior to the convective event. The time of the outlook was chosen to be the most recent update relative to the storm coming in. The SPC updates five times every day: 1

am, 8 am, 11:30 am, 3 pm, and 8 pm CST. For each day, the tornado probability assigned to Aurora was based on the most recent update prior to the onset of the storm.

SPC risk categorization: record tornado probability tiers (2%, 5%, 10%, 15%) from the outlook. Based on the map or, if Aurora's placement was ambiguous, "Some Larger Population Centers in Risk Area" section (though Aurora was not always large enough population-wise to be in this section). When Aurora's placement was ambiguous, a smaller-scale map released by NWS Lot was referenced.

SPC percentages were primarily treated as exact (e.g., 2%=0.02 exactly) in the primary analysis. They are treated this way because it is how the SPC conveys the percentages the majority of the time (it is quite hard to find public statements that indicate they are threshold). As seen in Figure 2, SPC risks are technically defined as being threshold ranges (as in 2% could be 2-4.99%), so further sensitivity analyses were conducted for threshold interpretations of forecast percentages.

Figure 2. Example of SPC threshold probabilities from a smaller-scale map released by NWS Chicago



(*Event Ready*, n.d.)

Tornado definition: Any level of tornado (EF-0–EF-5) reported by the NWS/SPC whose lat/long was within 25 miles of the selected Aurora point at any point in its journey.

Days where the percentage risk of tornadoes was listed as less than 2% but there was a tornado were collected but only used to determine baseline climatological frequency, as the SPC does not assign specific probabilistic bins to these dates, making consistent categorization difficult.

Distance method: measured using both the great circle distance from the listed tornado lat/long from the SPC report, and the reference Aurora point. Calculated using the NOAA great circle calculator and double-checked by Google Earth measuring tools. Note that the NOAA great circle calculator rounds to the nearest mile; if this were the case, then Google Earth was used as a backup. The SPC defines the forecast radius as 'within 25 miles'; accordingly, 25.00

miles was used as the inclusion threshold. Additional tests were done with a hypothetical 10-mile radius.

Multiple tornadoes occurring on the same day are only counted as one per the SPC tornado risk category definition.

Daily trials are treated as independent, though some temporal autocorrelation may exist during multi-day severe weather outbreaks (though multi-day outbreaks only happened once or twice through the whole 10-year span, and the majority of tornado days happened apart from one another). Temporal autocorrelation may result in slight overconfidence in p-values.

Lat/Longs of tornadoes are decided based on the listed lat/long of the tornado report; in rare cases, when a tornado was reported outside the region but close, an SPC storm path for the tornado would be referenced.

Used a conservative classification rule, where if Aurora sat on the physical border of the region (and the list of populated areas in said regions did not have Aurora because the cities in the list were all more populated), then it was assumed to be the lower percentage. Though this rarely appeared, and hence would have little effect on the study.

C. Data preparation

Collected from sources into a spreadsheet: with one row per day/outlook. Key columns included ‘date’, ‘year’, ‘SPC risk %’, ‘Hatched (Y/N)’, ‘Tornado ≤25Mi (Y/N)’, ‘Distance (Mi)’, ‘EF-rating’, ‘UTC of outlook if not 1630z’, ‘Time of tornado’, and ‘notes’. Not all data is relevant/used in this research.

Data grouped into full 2015-2025, 2015-2019, and 2020-2025.

Some data and statistical calculations may be rounded at certain points, typically to the hundredths place.

D. Tornado verification process

The first step in the tornado verification process was to review the SPC product archive every day from 2015 through 2025 and see if Aurora was in the risk boundary. If Aurora was in the risk category, SPC verification and SPC detailed storm reports would be consulted, and the lat/long of tornadoes that are close enough to be in the region (note: tornadoes in the verification clearly too far away, such as tornadoes in different states, would not be manually tested) would be calculated using the NOAA great circle calculator. If the tornado was within 25 miles, the

tornado would be counted as a ‘k’ day. Otherwise, there would be no tornadoes that day. The verification process was independently repeated an additional time to minimize human error. Data would be collected through the spreadsheet.

E. Statistical approach

1. Hypotheses

The null hypothesis is that tornado occurrence within 25 miles of the study location follows the SPC forecast probability for that category (i.e., the true probability of the event is p, as in the SPC is completely well-calibrated).

The alternative hypothesis is that the true probability is greater than p (i.e. SPC under-represents risk to the public, underforecast bias).

A forecast is perfectly calibrated when the observed frequency of events matches the forecast probability.

2. Metrics

Comparison metrics:

Observed proportion vs expected proportion

Ratio = observed/expected

3. Tests used

Binomial tests were used to compute the probability of observing k or more tornado days under a Binomial(n, p) model.

Because this study is specifically testing for underprediction:
Null Hypothesis: p=pSPC
Alternative Hypothesis: p>pSPC
Where p is in decimal form (e.g., 2% = .02)

Used Google Sheet formulas for the Wilson Interval.

Note: For higher SPC risks, there is a smaller sample size

Note that both the Wilson Confidence interval and the Binomial test p-value results were multiplied by 100 to achieve standard percentage values.

Relative risk is the observed tornado % on a given day divided by climatology. It comes out to be a multiplier (e.g., a relative risk of 15 for a 2% chance means that when SPC gives a 2% chance, it can be interpreted as approximately 15 times the relative increase of a tornado than the average day compared to climatological baseline frequency). Relative risk is a good indicator of the SPC's ability to distinguish elevated-risk days (also called risk stratification), which is not particularly related to a forecast being well-calibrated.

Climatology is the percentage of days on which there were tornadoes in a year. This can be figured out by taking the total number of tornado days (including days on which a tornado occurred at below 2%) and dividing it by the total number of days in the study (24/4018). This result turned out to be 0.60%.

F. Scale of data

A total of 4018 days were manually reviewed twice. There were 134 risk days (3.34%) and 24 tornado days (0.60%).

G. Definitions of Key Variables

Risk day counts: the number of days the SPC issued an SPC risk outlook for Aurora.

Expected counts: expected = n × p, where n = number of days with that SPC % and p = stated probability (e.g., 0.02 for 2%).

Each SPC probability category was analyzed separately under its corresponding binomial model.

Observed %: calculated by the formula k/n, where k is the number of days with a certain percent risk that has an observed tornado and n is the number of days with that SPC risk.

A 95% confidence interval provides a range of plausible values for the underlying population proportion. Under repeated sampling, 95% of confidence intervals constructed using this method would be expected to contain the true proportion. If the same sampling process were repeated many times, 95% of the resulting intervals would contain the true proportion. Wilson 95% interval is an alternative equation for this confidence interval that works better with smaller sample sizes, hence why it was chosen in this study. (*Confidence Intervals*, n.d.)

A p-value is the probability of observing results at least as extreme as those observed, assuming the null hypothesis is true (i.e., if the SPC was completely well-calibrated in their risk indication, what is the chance that the outcome happened). A right-tailed P-value binomial test is a test that asks the chances that the number of outcomes (k, observed tornadoes) or greater happens. If testing under-prediction, this is a superior method because it tests multiple outcomes (that number or higher) while simple P-value binomial tests the probability that that outcome, or an outcome more extreme, happens. (*Binomial Test*, n.d.)

Results:

Across all categories, the observed percentage of tornadoes was higher than the SPC's publicly conveyed percentage, particularly in the latter time period. This pattern was less apparent in the percent categories with a smaller sample size, however. An indication of an increased number of tornadoes than expected can be seen through both the right-tailed p-value binomials and the Wilson 95% Confidence intervals.

Overall Test results

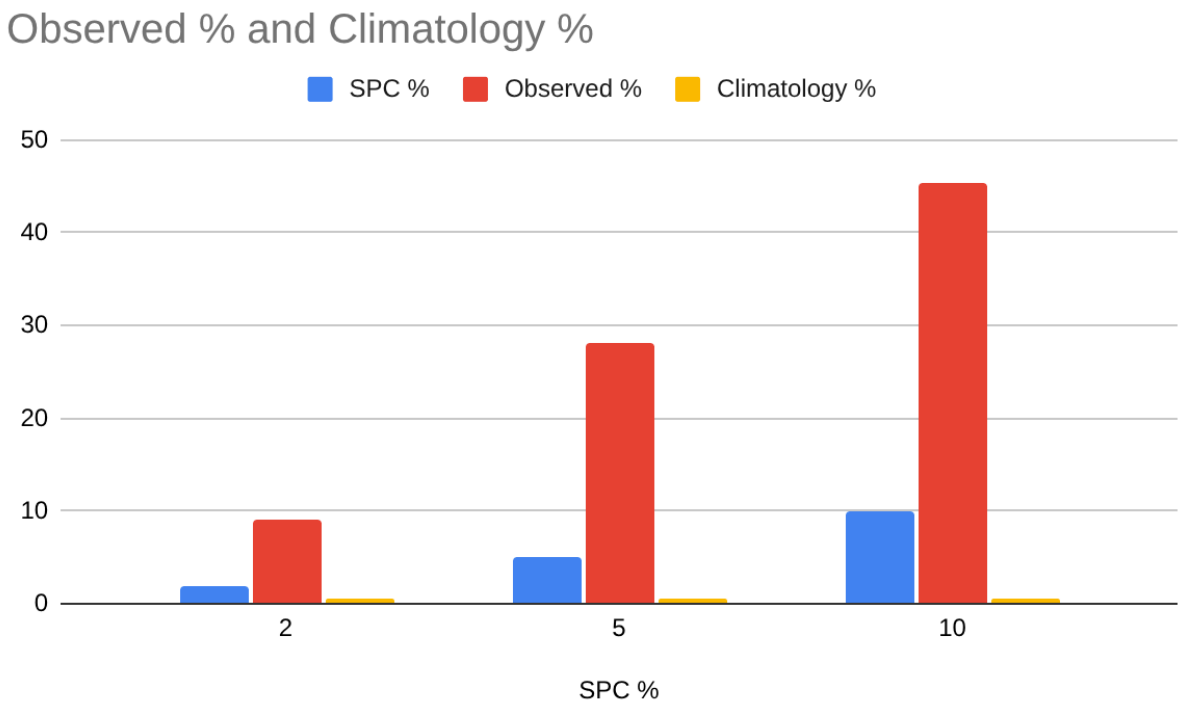
The overall results are an indication of the percentages taken without a threshold considered. This tracks both time periods and the entire span as well. This follows the SPC's guides and assumes the specified 25-mile radius. All categories are assessed, including 15% despite the low frequency of occurrence. However, the 15% category was retained descriptively for completeness but excluded from inferential interpretation because only one forecast occurred. Additionally, the 10% category was considered suggestive of a result but not absolute due to its low frequency.

Table 1. Overall Verification Results

SPC %	Time period	# of days (n)	expected # of tornadoes (n*p)	observed # tornadoes	observed %	ratio observed/ expected	Risk stratification (k%/c%)	right-tailed p-value binomial (%)	95% CI
2	All	89	1.78	8	8.99%	4.49	x14.98	0.042895869%	(4.60%-16.7%)
5	All	32	1.6	9	28.13%	5.625	x46.8	0.001911218%	(14.9%-45.3)
10	All	11	1.1	5	45.45%	4.55	x75.75	0.27509635%	(21.3%-71.9%)
15	All	1	0.15	0	0%	0	x0	100%	(0.00%-79.3%)
2	2015-2019	45	0.9	3	6.67%	3.33	x11.11	6.101435715%	(2.20%-17.8%)
5	2015-2019	8	0.4	0	0%	0	x0	100%	(0.00%-32.4%)
10	2015-	6	0.6	1	16.66%	1.67	x27.7	46.8559	(3.00%-56.35%)

	2019								
2	2020-2025	44	0.88	5	11.11%	5.68	x18.51	0.18157934%	(4.95%-23.9%)
5	2020-2025	24	1.2	9	37.50%	7.5	x62.5	0.000128315%	(21.0%-57.0%)
10	2020-2025	5	0.5	4	80.00%	8	x133.33	0.046%	(37.6%-96.4%)
15	2020-2025	1	0.15	0	0%	0	x0	100%	(0.00%-79.3%)

Figure 3. Observed % and Climatology % for overall results



Several trends are apparent in the overall results. Primarily, in samples with a larger number of instances, the observed number of tornadoes and the observed occurrence percent are much larger than the expected number of tornadoes. Under a binomial model assuming completely accurate SPC calibration, the P-value binomial tests show a low probability that the null hypothesis would occur, and the Wilson 95% Confidence interval additionally indicates discrepancies, both assuming perfect calibration from the SPC. This includes P-value binomial tests with statistically significant evidence against the null hypothesis and the fact that the SPC percentages are outside the Wilson 95% Confidence interval for many of the data rows. Additionally, it can be observed that throughout all the table's columns, these discrepancies between the SPC and observed risks are more prevalent through the 2020-2025 time period and

the all-time period. The 2015-2019 time period has fewer discrepancies and is even statistically consistent with the SPC in many places.

Assuming a medium threshold for SPC %'s.

The medium threshold data analysis is the data from the Overall Test reinterpreted to assume the SPC's risk is between either percent. In SPC risk percentages, 2% technically means 2-4.99%, as it is a threshold. This takes the average of the two extremes for a medium threshold. This test was done to analyze the tests from a middle baseline, as it can be more indicative of the intended variation within the SPC results, as it is an average between the two extremes. Note that this is still not particularly the main test, as that is done with the exact percentages (2, 5, 10, etc.) because those are the ones most publicly communicated. All statistical tests are recalculated, and 15% is taken out due to low sample size frequency. This test was done as a sensitivity analysis to test robustness and confirm that the data can withstand more conservative null hypotheses.

Table 2. Medium Threshold Sensitivity Analysis

SPC %	Time period	# of days (n)	expected # of tornadoes (n*p)	observed # tornadoes	observed %	ratio observed/expected	right tailed p value binomial	95% Cl
3.5	All	89	3.11	8	8.99%	2.57	1.289836665%	(4.6%-16.7%)
7.5	All	32	2.4	9	28.13%	3.75	0.04282143909%	(14.9%-45.3%)
12.5	All	11	1.375	5	45.45%	3.64	0.7330715889%	(21.3%-71.9%)
3.5	2015-2019	45	1.575	3	6.67%	1.9	20.82060867%	(2.2%-17.8%)
7.5	2015-2019	8	0.6	0	0%	0	100%	(0%-32.4%)
12.5	2015-2019	6	0.75	1	16.66%	1.377	55.12046814%	(3%-56.35%)
3.5	2020-2025	44	1.54	5	11.11%	3.25	1.835470464%	(4.95%-23.9%)
7.5	2020-2025	24	1.8	9	37.50%	5	0.00346142579%	(21.16%-57.3)
12.5	2020-2025	5	0.625	4	80.00%	6.4	0.1098632813%	(37.6%-96.4%)

As can be seen, due to the higher base probability, there is slightly less of a discrepancy when comparing the SPC percentages to the real results as compared to the base percentages. However, discrepancies still exist, particularly in the overall and 2020-2025 time periods. In the All and 2020-2025 time periods, it faces similar discrepancies, such as statistically low P-value percentages and the SPC percentages being outside the Wilson Confidence interval. The 2015-2019 timeframe, however, is even more consistent with SPC probabilities through those measures.

Assuming a high threshold for SPC %'s

This is the data from the Overall test reinterpreted to assume the SPC's risk is the highest it can be within the threshold. It ends in .99 as it cannot be the next number up, such that it needs to be in the same category. This test was done to ensure that the findings would be accurate even under the most conservative definition of the percentages. All statistical tests are recalculated, and 15% is taken out due to low sample size frequency. This test was done as a sensitivity analysis to test robustness and confirm that the data can withstand more conservative null hypotheses.

Table 3. High Threshold Sensitivity Analysis

SPC %	Time period	# of days (n)	expected # of tornadoes (n*p)	observed # tornadoes	observed %	ratio observed/ expected	right tailed p value binomial %	95% CI
4.99	All	89	4.441	8	8.99%	1.813	7.659118795%	(4.6%-16.7%)
9.99	All	32	3.193	9	28.13%	2.815	0.327306814%	(14.9%-45.3%)
14.99	All	11	1.64	5	45.45%	3.03	1.584438513%	(21.3%-71.9%)
4.99	2015-2019	45	2.245	3	6.67%	1.33	39.11055481%	(2.2%-17.8%)
9.99	2015-2019	8	0.799	0	0%	0	100%	(0%-32.4%)
14.99	2015-2019	6	0.899	1	16.66%	1.111	62.25841829%	(3%-56.35%)
4.99	2020-2025	44	2.195	5	11.11%	2.277	6.701742989%	(4.95%-23.9%)
9.99	2020-2025	24	2.39	9	37.50%	3.75	0.03188498601%	(21.16%-57.3%)
14.99	2020-2025	5	0.75	4	80.00%	5.34	0.2221767898%	(37.6%-96.4%)

Results for the high threshold appear similar to the medium, except results lean a small bit closer towards the SPC's predictions, with slightly lower discrepancy. There is still a statistically significant discrepancy in the 2020-2025 time period in measures such as the P-value binomials and the Wilson 95% Confidence interval, and additional discrepancies in the 'All' time period; however, these results are marginally less prevalent overall.

Assuming a 10-mile radius

This is the data reinterpreted and calibrated to assume a 10-mile radius instead of the standard 25-mile one. This test was performed as a sensitivity analysis to find the SPC's sensitivity to spatial scale and test robustness. 15% is taken out due to low frequency.

Table 4. 10-Mile Radius Sensitivity Analysis

SPC %	Time period	# of days (n)	expected # of tornadoes (n*p)	observed # tornadoes	observed %	ratio observed/ expected	right tailed p value binomial (%)	95% CI
2	All	89	1.78	1	1.12%	0.56	53.35514547%	(0.19%-6%)
5	All	32	1.6	2	6.25%	1.25	21.38855271%	(1.7%-20.1%)
10	All	11	1.1	3	27.27%	2.27	1.85347612%	(9.7%-56.6%)
2	2015-2019	45	0.9	0	0.00%	0	100%	(0%-7.8%)
5	2015-2019	8	0.4	0	0.00%	0	100%	(0%-32.4%)
10	2015-2019	6	0.6	0	0.00%	0	100%	(0%-39%)
2	2020-2025	44	0.88	1	2.22%	1.13	22.71322604%	(0.4%-11%)
5	2020-2025	24	1.2	2	8.33%	1.67	11.59445884%	(2.3%-25.8%)
10	2020-2025	5	0.5	3	60.00%	6	0.046%	(23%-88.2%)

Results for the 10-mile radius become closer to expected based on the communicated SPC percentages, with the data for all the time periods containing fewer discrepancies when compared to the SPC stated percentages. Tests such as the P-value binomial and the Wilson 95% Confidence Interval show data that match the SPC's results. The only exception to this is the 10% All and 2020-2025 time period, which still shows major discrepancies. However, there is a small sample size of just 5 days for the 10%.

Relative Risk

This table marks the relative risk, which is the variation between the observed findings for each SPC risk percent and baseline climatology for each SPC category.

Table 5. Risk Stratification Compared to Climatology

SPC %	Observed %	Climatology %	Relative Risk

2	8.99	0.6	x14.98
5	28.13	0.6	x46.8
10	45.45	0.6	x75.75

Risk stratification is not a percentage; instead, it is a multiplier that indicates how many times higher the risk is than the average day. Using climatology, a daily percentage of 0.6% of a tornado on any given day is concluded (climatology, in this usage, does not take into account the prominent tornado seasons and thus is not a perfect indicator). From this climatology, the relative risk is an indication of (when the SPC releases a given tornado percent risk) the amount, in multiples, by which the risk is higher that day than the climatology baseline. The risk stratification is quite high, meaning on days in which the SPC releases a risk category, regardless of whether the percentages are in agreement with observed frequency, there is an increased likelihood of a tornado occurring.

Discussion

The results suggested that risk forecasts in a local area ended up having poor calibration, particularly in the 2020-2025 data set, if they are tested strictly as publicly communicated. Overall, the observed percentage of tornadoes tended to be higher than the SPC assessment, which indicates over-conservatism and an underforecast bias when reporting. This finding of an underforecast bias is consistent with the findings from Gregory R. Herman, Erik R. Nielsen, and Russ S. Schumacher's report on the national scale and with proper meteorological tests. For many of the observed percentages, their Wilson 95% confidence intervals did not, or only barely, include the SPC's assessment in them, again indicating poor locally communicated calibration. Additionally, for the overall and the 2020-2025 period, the binomial test p-values also suggested poor local calibration for the SPC. These results and findings stayed generally consistent at higher thresholds, which suggests that the findings are able to withstand some conservativeness with higher assumed percentages.

It is not ideal that the SPC yielded poor local calibration, as many people treat these numbers as telling the full story. Although in reality, the SPC percentages are only one part of a larger meteorological picture (as can be seen through the fact that such national verifications undergo complex tests), this is not readily expressed to the general public. Additionally, there are questions about whether it is reasonable to expect there to be high local calibration in the first place, even if the data was tested using strict meteorological verification methods. Within any given location, there is the capability of natural shifts and variation that may cause one area to have slightly higher discrepancies. For many forecasts, the focus is on evening out results in the long term, as even with certain regions being poorly calibrated, the results can become better

calibrated when taking the national results as a whole. It is possible that a nationally well-calibrated forecast can be locally miscalibrated, as the local discrepancies would even out on the national scale. While it is ultimately ideal that any given point verifies statistically, that will not be the case for many.

One speculation could be that the poor calibration in publicly communicated results could stem from an emphasis on predicted larger tornadoes. While this is not confirmed, it could make sense, as weak QLCS tornadoes would likely skew the percentages statistically significantly higher, causing more public panic. If QLCS tornadoes are treated as normal tornadoes, a significant QLCS day could have higher natural tornado probability as an actual 'high risk day' with meaningful chances of producing destructive tornadoes. Because of that, it would make sense if the SPC were to decide to make a distinction between these small spin-ups and meaningful tornadoes. This would make sense given that nearly all of the tornadoes were under EF-2 and a good portion of them were QLCS spin-ups. Whether or not this is an intentional choice made by the SPC, their public communication does not reflect a distinction, which would leave the public without the context to interpret the risk correctly. However, any evidence of this is outside the scope of this study, but would be interesting to test under future research.

It is also possible, though highly speculative, that the SPC creates the forecasts with nuances but believes that it is too technically advanced for the general public, so they do not release it. While there is no evidence for this other than the local probabilities having poor calibration, this would be problematic as it is also limited to many members of the general public who might have the ability to interpret it despite not being professional. In general, it is likely better than all information and nuances are put out to the general public, even if it is technically advanced.

In a more general stance, it is also possible that, due to the scale of the SPC, perfect local calibration at any given point may not be realistic, as they do not have the resources to emphasize this with that much detail, or the nature of the events is too unpredictable to accurately gauge the risk for any given town. However, it could be possible that these discrepancies, particularly for the 5% and 10%, are outside of the natural variation of what is expected with accurately publicly conveyed forecasts.

Further questions arise as to why there is a large shift from the 2015-2019 time period to the 2020 time period. This could be a natural shift in overall SPC calibration, or a shift in how SPC decides to communicate their numbers. One possible natural speculation as to why is simply that tornado detection has significantly increased throughout the years. With the advancement of technology, such as possible radar improvements or even AI starting to be introduced in detection (Foy, 2024), among many other advancements, tornadoes may have an easier time being detected. Additionally, the number of tornado chasers has likely risen over the years due to

the internet, which may also lead to higher detection of weaker tornadoes, as previously they may not have been as noticeable through radar due to their smaller size and lesser debris (strong tornadoes are typically more apparent in the first place and are not as affected by this).

Higher detection would naturally lead to an artificial ‘increase’ in tornadoes, where it appears as if there are more tornadoes when the reality is that the true number of tornadoes is similar. It could be speculated that the SPC risks have not fully adjusted to this rapid shift and are still basing their models on a time period with fewer tornadoes. This idea can answer some questions, but also raises a few. It makes sense with the results in the sense that most of Aurora’s tornadoes within the time period were under EF-2 (23 of the 24), which would match the hypothesis that there are increased detection abilities. However, it raises some questions, such as the rapid shift. While it is true that technology today is better than it was in 2019, advancement has likely not outpaced SPC predictions so much and so quickly that it would lead to these results. Thus, it is not likely that this is the entire explanation for the discrepancy, rather one part of a larger picture of multiple factors that cause low calibration.

Another possible conclusion is simply natural variation in the number of tornadoes. There were, during the 2020-2025 time period, more reported tornadoes in northern Illinois as a whole. In recent years, Illinois has led the nation in the number of tornadoes, having the most tornadoes in 2023, 2025, and the second most in 2024 (Miller & Yeomans, 2026). The seemingly quick shift of tornado frequency towards Illinois in recent years may contribute to the low calibration from the SPC, as the number of tornadoes may be considered to be unprecedented or outside the typical range. A possible reason for the poor calibration may be that Illinois has been exceeding historical averages in general and exceeding the expected climatology, or it is also possible that the SPC simply had not updated its forecasts to deal with the rise of tornadoes in Illinois, though this remains speculative. Of course, it can also be mentioned that, with any probabilistic event, there is also a chance that Aurora has just ‘beat the odds’ in some way. While it is likely not the main factor, the inherent randomness has likely played some role, even if a small one.

Interestingly, the 10-mile radius yielded results that are much more consistent with the SPC. The 25-mile radius is quite large, and this was done as it is “approximately the size of a major metropolitan area.” With an underforecast bias, it is likely that the more constrained radius would lead to fewer tornadoes, which would help mitigate such a bias by having less area for tornadoes. In this specific trial, the 10-mile radius would have led to better calibration; however, the 10-mile radius is ultimately quite arbitrary and not particularly conclusive. For example, all three categories, the 2%, 5%, and 10%, had no tornadoes at all for the 2015-2019 time period, as seen in Table 4, which significantly raises limitations and suggests it could just as easily have been a sample-size artifact. The SPC uses a 25-mile radius to reflect the size of a major metropolitan area, and thus, a change to a 10-mile radius may be unideal, as it may not be as representative for someone living in a major metropolitan area.

Although the SPC local calibration tended to be low overall, it had high risk stratification in its forecasts. Climatology across the 11 years was a 0.60% chance on any given day that a tornado would occur (Wilson CI: (0.40-0.89%)). Given that observed percentages are significantly higher than climatology across all risk levels, it can be indicated that the SPC had high risk stratification for Aurora, as they were able to correctly identify days in which there was a higher risk of an event occurring, even if the forecast had poor calibration in terms of the percentages. Throughout the period, there were only 2 tornado days in which there was no tornado forecast ('under 2%,' meaning that it is still possible for there to be a tornado), while throughout the entire period, there were 22 tornado days in total under SPC risks. These results indicate that the SPC can accurately indicate which days will be of higher risk to produce a tornado.

Assuming these results are correct and that SPC calibration is low and risk stratification is high, what would this mean for someone who is living in Aurora? It would mean that, in recent years, it may be more useful to think in terms of relative risk, as that is what has appeared to provide a better indication through the high risk stratification when compared to the poor percentile calibration. When one sees 10%, for example, instead of thinking purely in raw numbers, it may be better just to think 'the probability of seeing a tornado today is much higher than usual, so I should be alert.' These findings suggest that these days may warrant more attention than the number particularly suggests. 5% sounds quite low to the average person, yet it is an indication of a real tornado risk, and it is important that the public understands this and does not completely neglect the risk.

In fact, this is what many other weather services already do, as many non-government weather providers take SPC's tornado probability and directly recalculate on an arbitrary scale. For example, on The Weather Channel, they operate on a scale of 1 to 5, with 5 being the highest odds of a tornado, equivalent to a 30% SPC category (note this is different from 'Torcon', which recalculates and differs from the SPC). These labels are directly taken from the SPC, but communicated to the public in a way without specific percentages (Belles & Dolce, 2025). The SPC already does this in some way, with their 'marginal, slight, enhance, moderate, high' categories; however, this is separate from their tornado probabilities. Overall, this causes the public to be aware of the general risk, without particularly assigning a percentage. It may be easy for a member of the public to dismiss that 5% risk, as it sounds low, but when faced with an arbitrary tornado risk which is higher than many days, it may cause attention.

These results do not particularly indicate anything about national results, and cannot be made to serve any national trend in themselves, as one city cannot establish national conclusions. However, it may be beneficial to see a widespread finding for the public's perspective verification in the future. Most SPC national calibration is based on widespread statistics, yet it

could also be useful to conduct these calibration tests for many cities, as it can also be useful for the SPC to ensure better calibration in the way that the forecasts are meant to be used.

Limitations

Given the nature of the one-city sample size, it is clear that the biggest limitation is the low sample size. This is a fundamental limitation that is apparent in the scope. Tornadoes themselves are quite rare events for any given region, and there is an overall low sample size because of this. Additionally, a low sample size may lead to a higher uncertainty of results, as slight variations can substantially affect the results. It is possible that one city simply has more tornadoes, as the SPC risk categories are more broadly drawn. By the nature of the study, there was nothing that could be done with this flaw. One single point was chosen in Aurora, which is necessary for consistent verification but could also lead to some slight variation even within this city.

The SPC tornado reporting is also not a perfect measure of tornado frequency, as the listed coordinates can vary in exact location and overall path. Daily observations were also treated as independent despite potential autocorrelation during outbreaks.

An additional limitation is that the SPC categories were manually checked by one individual, which may lead to human error. To minimize the risk of human error, data were manually checked twice for each date, and a third, quicker check for confirmation was performed. However, it is still possible that there are some errors or oversights because of this.

No correction for multiple comparisons was applied, given the exploratory nature of the study.

For simplicity purposes, the relative risk and daily tornado climatology were assumed to be consistent throughout the year; however, in practice, certain time periods throughout the year have higher daily tornado risk according to climatology.

For research purposes, specific SPC threshold values had to be assumed, and for each assumption, all points had to be treated as if they were the same percentage value. This was done as it best matches how the percentages are publicly communicated, but is a limitation in a broad verification sense.

Conclusion

Were local SPC Day-1 probabilities well-calibrated at Aurora, IL, between 2015 and 2025 if measured by public communication?

Overall, local observed tornado frequencies exceeded SPC percentages and indicate an underforecast bias, most evident in the 2020-2025 time period. Findings remained consistent even under more conservative threshold assumptions. Higher risk categories generally followed the same trend, though their lower sample size limited confidence in those findings.

While local calibration was weaker than expected, SPC forecast discrimination remained high. The SPC was still able to identify days of risk, and the forecasts can still be useful even if agreement with observed frequency appears to be lower than expected.

Did the observed relationship between SPC probability categories and tornado occurrence differ between the two periods?

The 2015-2019 results were consistent with the stated probabilities, while the 2020-2025 time period had results that yielded findings of poor calibration. As such, there was a difference in the occurrence of each percentage over the period. However, the small number of events prevents a strong conclusion that the calibration itself changed statistically.

These findings matter because SPC forecasts are consumed locally and at face value, and an underforecast bias might lead to the public not taking these results as seriously as they should. However, a forecast can still be useful even when calibration is imperfect, because of the high skill that indicates important information.

A local calibration of the SPC can yield useful results that are not immediately apparent in national tests. It is also important to do additional local calibrations to determine if Aurora's results are unusual or representative of what could be expected from many different cities. More accurate conclusions can be drawn by performing this local verification on a national scale using many different cities. It would also be beneficial to see how SPC forecasts hold up from the public's perspective on a national level, and more research can be done.

The point of this study was not to test the SPC's forecast in a strict meteorological way; instead, it was to test if their presentation and language can hold up from the perspective of the general public. In all, this study found that the SPC's values may not particularly hold up when taken from the public's perspective. As such, it would likely be beneficial if there is more communication made to the public over how the SPC creates its forecasts and what it means.

Overall, the SPC's communication of its risk should be as transparent as possible. If their forecasts contain any nuance, it is important that this is publicly conveyed, as it gives the public the best ability to understand the situation and the risk and to make the appropriate choices. As seen in the study, the publicly communicated probabilities do not always hold up if tested strictly

through the SPC's public framework of treating each event as independent statistical occurrences. A percentage without context implies a straightforward probability, and thus the public will treat it as such.

References

Belles, J., & Dolce, C. (2025, March 14). *What Severe Thunderstorm Outlooks Mean For The Tornado, Wind Damage And Large Hail Threat Level*. Weather.com. https://weather.com/storms/severe/news/2025-03-14-thunderstorm-tornado-outlooks-storm-prediction-center

Binomial Test. (n.d.). University of Texas at Austin. https://sites.utexas.edu/sos/guided/inferential/categorical/univariate/binomial/

Bunting, B., & Grams, J. (2020, February 29). *SPC Convective Outlooks*. Storm Prediction Center. https://www.spc.noaa.gov/misc/SPC_probotlk_info.html

Carney, K. (2026, February 2). *Illinois Cities by Population (2026)*. Illinois Demographics. https://www.illinois-demographics.com/cities_by_population

Confidence intervals. (n.d.). NIST. https://www.itl.nist.gov/div898/handbook/prc/section2/prc241.htm

Event Ready. (n.d.). National Weather Service. https://www.weather.gov/crh/eventready?sid=lot

Foy, K. (2024, April 16). *An AI dataset carves new paths to tornado detection*. MIT. https://www.ll.mit.edu/news/ai-dataset-carves-new-paths-tornado-detection

Google Earth. (n.d.). Google Earth. https://earth.google.com/web/

Latitude/Longitude Distance Calculator. (n.d.). National Hurricane Center and Central Pacific Hurricane Center. https://www.nhc.noaa.gov/gccalc.shtml

Miller, K., & Yeomans, D. (2026, April 23). Illinois leads the nation in tornadoes for 2026, with confirmed count nearly 100. *CBS*.

https://www.cbsnews.com/chicago/news/illinois-tornado-count-2026-most-in-country/#:~:text=In%202023%20and%202025%2C%20Illinois,year%20with%20153%20confirmed%20tornadoes

Schumacher, R. S., Nielsen, E. R., & Herman, G. R. (2018, February 1). Probabilistic Verification of Storm Prediction Center Convective Outlooks. *Weather and Forecasting*, *33*(1), 161-184. https://journals.ametsoc.org/view/journals/wefo/33/1/waf-d-17-0104_1.xml

SPC Convective Outlooks: Days 1–8. (2026, March 2). Storm Prediction Center. https://spc.noaa.gov/about/outlooks/#probToCat

Storm Prediction Center Product & Report Archives. (n.d.). Storm Prediction Center. https://www.spc.noaa.gov/archive/

Storm Prediction Center Product & Report Archives. (2020, March 28). Storm Prediction Center. https://www.spc.noaa.gov/products/outlook/archive/2020/day1otlk_20200328_1630.html

Storm Reports page. (n.d.). Storm Prediction Center. https://www.spc.noaa.gov/climo/reports/